



PROJECT SEVEN MILE
INSPECTION AND TEST PLAN SECTIONS

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1	100	Receiving	00	01/08/13
2	200	Draft tube liner elbow	00	01/08/13
3	300	Draft tube liner cone	00	01/08/13
4	400	Piezometer and miscellaneous embedded piping	01	01/10/17
5	500	Discharge ring	01	02/01/21
6	600	Stay ring	02	03/01/14
7	700	Spiral casing	00	02/01/10
8	800	Pit liner	00	02/03/14
9	900	Servomotor mounting plates	00	02/03/14
10	1000	Machining	00	02/03/14
11	1100	Bottom ring	01	02/10/17
12	1200	Head cover and wicket gates	01	02/10/17
13	1300	Runner and turbine shaft	01	02/11/05
14	1400	Wicket gates operating mechanism	00	02/11/25
15	1500	Servomotors	00	02/12/13
16	1600	Shaft seal assembly	00	02/11/25
17	1700	Turbine guide bearing	00	03/01/10
18	1800	Air admission	00	03/01/13
19	1900	Governor system	00	03/01/13
20	2000	Foundation	00	02/07/11
21	2100	Stator	00	02/07/11
22	2200	Rotor	00	02/07/18
23	2300	Lower bracket, thrust and guide bearing	00	02/09/19
24	2400	Upper bracket	00	02/12/06
25	2500	Unit alignment and rotation	00	02/09/19
26	2600	Piping (turbine and generator)	00	02/12/10
27	2700	Instrumentation and Pre-commissioning (T/G)		
28	2800	Generator bearing modifications	00	03/03/27

INSPECTION AND TEST PLAN

Section 1500

SERVOMOTORS

Project: SEVEN MILE

Unit 4



GE Hydro

INSPECTION AND TEST PLAN

Project SEVEN MILE

Customer: B.C. Hydro

Contract: SM-70

Unit 4

Section 1500: Servomotors

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1501	100	I,R	Incoming inspection.	See ITP Section 100		QCR 101	
1502	1900	O	Ensure all ports are hermetically closed with blank flanges or plugs. Remove the pipe blank flanges placed on the servomotor ports. Drain the oil from the servomotors. Prepare the blank flanges to be able to use air to move the piston rods. Clean the two servomotor mounting plates in the pit liner. Mark the mounting plates.	Installation proc. SMI-U04-02056 Steps 2.1 to 2.6 Drawings 222038228 222038360 222038370			 Jan 11 th 2003
1503	1900	O	Install the studs into the mounting plates. Rig and lower the servomotors in the pit and mount the servomotors on their corresponding mounting plates. Bolt temporary.	Installation proc. SMI-U04-02056 Steps 2.6 and 2.7 Drawing 222038228			 Jan 11 th 2003

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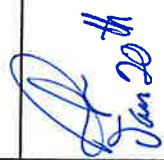
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1504	1900	O	Check and adjust the servomotor alignment, level and elevation.	Installation proc. SMI-U04-02056 Steps 2.8 to 2.10 Drawing 222038228	Shim as required		 Jan 20 th 2003
1505	1900	I,R	Dimensional inspection: -Servomotors level and elevation -Servomotors stroke -Servomotors alignment -Shim thickness and location	Installation proc. SMI-U04-02056 Steps 2.10 and 2.11 Drawing 222038228	See final QCR's ITP 1508	QCR 1901 QCR 1903 QCR 1905 QCR 004	 Jan 20 th 2003
1506	1900	O	Machine the field adjustment plates.	Installation proc. SMI-U04-02056 Step 2.12	If required	QCR 002 <i>Not Indicated</i>	 Jan 20 th

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				Drawing 222038228			<i>2003</i>
1507	1900	O	Mount the servomotors and their respective field alignment plate on to the mounting plates. Bolt temporary.	Installation proc. SMI-U04-02056 Step 2.13 Drawing 222038228	If machining is necessary		<i>2003</i> <i>Jan 21st</i>
1508	1900	O	Dimensional inspection: -Servomotors level and elevation -Servomotors stroke -Servomotors alignment	Installation proc. SMI-U04-02056 Step 2.13 Drawing 222038228	If machining is necessary	QCR 1901 QCR 1903 QCR 1905	<i>2003</i> <i>Jan 23rd</i>
1509	1900	O,M,R	Install the remaining studs. Remove and clean the first 8 studs. Re-install them. Torque.	Installation proc. SMI-U04-02056 Step 2.13	Torque value is 1520 N.m lub.	QCR 004 ✓	<i>2003</i> <i>Feb 12th</i>
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			Torque monitoring.	Drawing 222038228			
1510	1900	O	Install the two dowel pins per servomotors.	Installation proc. SMI-U04-02056 Step 2.14 Drawing 222038228			<i>[Signature]</i> Feb 9 th 2003
1511	1900	O	Extend the rod end fully and check the thrust washer seat elevation with the gate operating ring. Machine one of the two thrust washer if required.	Installation proc. SMI-U04-02056 Step 2.15 Drawing 222038228			<i>[Signature]</i> Jan 2 nd 2003
1512	1900	I,R	Dimensional inspection: -Servomotors level and elevation	Installation proc. SMI-U04-02056 Step 2.15 Drawing		QCR 1901 <i>Not done</i>	<i>[Signature]</i> Jan 2 nd 2003

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				222038228			
1513	1900	O	Clean the servomotors connecting rods and install on the gate operating ring only.	Installation proc. SMI-U04-02056 Step 2.16 Drawing 222038228			 Jan 25 th 2003
1514	1900	O	Pull out the servomotor piston rod #2 until piston contacts servomotor head, and inversely, push in servomotor piston rod #1.	Installation proc. SMI-U04-02056 Step 2.17 Drawings 222038360 222038370			 Jan 25 th 2003
1515	1900	I,R	Measure distance from servomotor #1 head to the rod end.	Installation proc. SMI-U04-02056 Step 2.17		QCR 1908	 03/00/03

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							initials
1516	1900	O	Move in servomotor #2 piston 16 mm +/- 1 from the cylinder head and move out servomotor #1 piston 16 mm +/- 1 from the back plate.	Drawings 222038360 222038370			—
				Installation proc. SMI-U04-02056 Step 2.18			<i>[Signature]</i> Jan 23 rd 2003
1517	1900	I,R	Measure distance from servomotor #1 head to the rod end.	Drawings 222038360 222038370		QCR 1908	<i>[Signature]</i> MAR 10 / 03
				Installation proc. SMI-U04-02056 Step 2.18			
1518	1900	O	Couple the both servomotors to the operating ring. Tighten snug the Supernut screws.	Drawings 222038360 222038370			—
				Installation proc. SMI-U04-02056			

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GE Hydro

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			Tighten snug the Supernut screws.	Steps 2.19 and 2.20 Drawings 222038360 222038370			<i>Feb 29th 2003</i>
1519	1900	O	Remove the air hoses and blank flanges. Connect the hydraulic piping to the servomotors.	Installation proc. SMI-U04-02056 Step 2.21 Drawing 222038288			<i>Feb 6th 2003</i>
1520	1900	O,M,R	Remove the shims installed between the gate operating ring and the keeper ring segments. Mount the keeper ring segment. Torque. Torque monitoring.	Installation proc. SMI-U04-02056 Step 2.22 Drawing 222038211	Torque value is 145 N.m lub.	QCR 004 ✓ <i>See ITP Step 1406</i>	<i>Jan 28th 2003</i>

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1521	1900	O	Open the wicket gates at maximum opening with hydraulic pressure and release pressure.	Installation proc. SMI-U04-02056 Step 3.1 Drawing 222038228			
1522	1900	I,R	Measure distance from servomotor head to the rod end. Measure the gap between the levers and the stopping blocks on the opening side.	Installation proc. SMI-U04-02056 Step 3.1 Drawings 222038226 222038360 222038370		QCR 1903, QCR 1810✓	 
1523	1900	O	Close the wicket gates, squeeze and release pressure. <i>check for intake MARKTO LINE UP - Gon - H/C.</i>	Installation proc. SMI-U04-02056 Step 3.2 Drawing 222038228		QCR 004	 <i>MAR 11/03</i>

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1524	1900	I,R	Measure distance from servomotor head to the rod end. Measure the gap between the levers and the stopping blocks on the closing side.	Installation proc. SMI-U04-02056 Step 3.2 Drawings 222038226 222038360 222038370		QCR 1903 QCR 1810	 MAR 1/03
1525	1900	O	Grind the stopping blocks on the opening side and install the Fabreeka pads.	Installation proc. SMI-U04-02056 Step 3.2 Drawing 222038226			 Feb 3rd 2003
1526	1900	O	Apply 10% of the maximum system pressure on the closing side.	Installation proc. SMI-U04-02056 Step 3.3 Drawing 222038228			 Feb 28/03

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1527	1900	I,R	Measure the distance between the rod end and the servomotor head on servomotor #1.	Installation proc. SMI-U04-02056 Step 3.3 Drawing 222038360		QCR 1908	 MAR/01/03
1528	1900	O	Apply 100% of the maximum system pressure on the closing side.	Installation proc. SMI-U04-02056 Step 3.4 Drawing 222038228			 Feb 28/03
1529	1900	I,R	Measure the distance between the rod end and the servomotor head on servomotor #1. Measure the wicket gates contact points	Installation proc. SMI-U04-02056 Step 3.4 Drawings 222038226	Full Sequence	QCR 1908 QCR 1802	 March 21/03

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1530	1900	I,R	Determine the elasticity of the mechanism. Measure the distance between the rod end and the piston rod or a reference point on the piston rod. Determine the required distance between the rod end and the piston rod on servomotor #1.	222038360 Installation proc. SMI-U04-02056 Steps 3.5 to 3.8 Drawings 222038360 222038370	QCR 1908	 MAR 1/03
1531	1900	O,M,R	Adjust the distance between servomotor #1 head and rod end or a reference point on the piston rod. Torque the Supernut. Torque monitoring.	Installation proc. SMI-U04-02056 Step 3.9 Drawing 222038360	QCR 004 Torque value is 380 N.m lub.	 Feb 28/03
1532	1900	I,R	Record the final distance between the servomotor head and the rod end.	Installation proc. SMI-U04-02056 Step 3.9 Drawing	QCR 1908 QCR 1903	 MAR 1/03

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				222038360			
1533	1900	O	Apply 100% of the maximum pressure on the closing side and check that the lock key can be engaged. Release pressure. Check that a squeeze remains on the wicket gates.	Installation proc. SMI-U04-02056 Step 3.9 Drawing 222038360	The lock key engage		 Feb 8 th 2003
1534	1900	O	Apply 100% of the maximum system pressure on the closing side and disengage the lock key. Open the wicket gates to open position. Release pressure. Check that the lock key on servomotor #2 can be engaged. Adjust shim thickness as required.	Installation proc. SMI-U04-02056 Step 3.10 Drawings 222038360 222038370	The lock key on SM#2 engage.		 Feb 8 th 2003
1535	1900	O	Grind the stopping blocks on the closing side and install the Fabreeka pads.	Installation proc. SMI-U04-02056 Step 3.11			

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1536	1900	I,R	Measure distance from servomotor #2 head to rod end.	Drawing 222038226			 Feb 5 th 2003
1537	1900	O	Close the wicket gates then release pressure.	Installation proc. SMI-U04-02056 Step 3.12 Drawing 222038370		QCR 1903	 MAR 1/03
1538	1900	I,R	Measure distance from servomotor #2 head to rod end. Record the stroke.	Installation proc. SMI-U04-02056 Step 3.12 Drawing 222038228			 Feb 28/03
				Installation proc. SMI-U04-02056 Step 3.12		QCR 1903	

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				Drawing 222038370			<i>Feb 2003</i>
1539	1900	O,M,R	Torque the Supernut on the servomotor #2. Torque monitoring.	Installation proc. SMI-U04-02056 Step 3.12 Drawing 222038370	Torque value is 380 N.m lub.	QCR 004✓	<i>Feb 28/03</i>
1540	1900	O	Install the necessary brackets for the transducer and switches for the governor system. Adjust positions. Install and adjust the stroke indicator on servomotor #1. Connect the piping to the actuator of the locking device. Once the governor system is functional, check the operation of the automatic servomotor locking device.	Installation proc. SMI-U04-02056 Steps 3.13, 4.1 and 4.2 Drawing 222038360			<i>Feb 20th 2003</i>
1540	1900	O	Retract servomotor #2 completely to open the wicket gates and lock the servomotor.	Installation proc. SMI-U04-02056			<i>✓</i>

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1541	1900	I,R	Measure the distance between the servomotor head and the rod end.	Step 4.3.1 Drawing 222038370			<i>[Signature]</i> 28 Feb 03
1542	1900	O	Machine the spacer (item 75) to measured dimension. Install the spacer.	Installation proc. SMI-U04-02056 Step 4.3.1 Drawing 222038370			<i>[Signature]</i> MAR/1/03
1543	1900	I,R	Final inspection :	Installation proc. SMI-U04-02056 Step 4.3.1 Drawing 222038370	No Machining Required.	QCR 004	<i>[Signature]</i> Feb 28th 2003
				Installation proc.		QCR 003	<i>[Signature]</i>

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			Ensure that all the work and documentation have been completed.	SMI-U04-02056			<i>[Signature]</i> 04/28/03

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